1. Consider an orthonormal basis $B = \left\langle \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \right\rangle$ for $\mathbf{R}^3$. Write a vector $\mathbf{v} = \begin{bmatrix} 1 \\ -4 \\ 7 \end{bmatrix}$ as the sum of basis B. (In the form of $\mathbf{v} = a \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} + b \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix} + c \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}, \exists a, b, c \in \mathbf{R}$)

2. Which linear transformations below are distortion-free? Justify your answer.

   2.1. $T\colon \mathbf{R}^3 \Rightarrow \mathbf{R}^3$, given by $T(x, y, z) = (z, x, y)$

   2.2. $S\colon \mathbf{R}^2 \Rightarrow \mathbf{R}^4$, given by $S(x, y) = \frac{1}{2}(x + y, x - y, x + y)$

3. Find an orthonormal basis from the condition in each question.

3.1. An orthonormal basis for the plane spanned by (1, 2, 3) and (3, 2, 1) in $\mathbf{R}^3$

3.2. An orthonormal basis for the nullspace of $\mathbf{A} = \begin{bmatrix} 1 & 6 & 5 & -2 \\ 3 & 2 & 7 & -2 \end{bmatrix}$

3.3. An orthonormal basis that spans the same subspace as the set $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ -1 \end{bmatrix} \right\}$

4. Consider a linear map $f\colon \mathbf{R}^2 \Rightarrow \mathbf{R}^3$ where $f\left(\begin{bmatrix}1\\-1\end{bmatrix}\right) = \begin{bmatrix}1\\-1\\2\end{bmatrix}$ and $f\left(\begin{bmatrix}2\\-3\end{bmatrix}\right) = \begin{bmatrix}3\\-7\\4\end{bmatrix}$.

4.1. If a basis $B = \left\langle \begin{bmatrix}1\\-1\end{bmatrix}, \begin{bmatrix}2\\-3\end{bmatrix} \right\rangle$, and $f(x) = (\mathbf{A}_{3\times 2})\big(\mathrm{Rep}_B(x)\big)$. Find $\mathbf{A}_{3\times 2}$.

4.2. Find $(a,\, b)$ such that $\begin{bmatrix}7\\-9\end{bmatrix} = a\begin{bmatrix}1\\-1\end{bmatrix} + b\begin{bmatrix}2\\-3\end{bmatrix}$.

4.3. Evaluate $f\left(\begin{bmatrix}7\\-9\end{bmatrix}\right)$.

4.4. Fill in the blank.

$\mathrm{Rep}_B(x)$ changes x representation from the _____ to the basis _____.
Equivalently, $\mathrm{Rep}_B(x)$ can be substituted with $\mathrm{Rep}_{____,B}(id(x))$.

We can say that $\mathbf{A}_{3\times 2}$ is equivalent to $\mathrm{Rep}_{_____,_____}(f)$.

As $\mathbf{A}_{3\times 2}$ performs a linear mapping _____ to the vector with the basis _____ representation and changes it to the _____ basis representation.

5. Consider a linear map $f\colon \mathbf{R}^2 \Rightarrow \mathbf{R}^3$ where $f\left(\begin{bmatrix} a \\ b \end{bmatrix}\right) = \begin{bmatrix} a+b \\ 2a-3b \\ a-b \end{bmatrix}$ and a basis $B = \left\langle \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \end{bmatrix} \right\rangle$

5.1. Find a matrix T such that $f(x) = Tx$ for all $\mathbf{x} \in \mathbf{R}^n$

5.2. Find $a_1 = f\left(\begin{bmatrix} 1 \\ -2 \end{bmatrix}\right)$ and $a_2 = f\left(\begin{bmatrix} -2 \\ 3 \end{bmatrix}\right)$. Find a_1 and a_2.

5.3. Find $\mathrm{Rep}_B(x)$ for all $x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$.

5.4. Is it true that $Tx = \begin{bmatrix} | & | \\ a_1 & a_2 \\ | & | \end{bmatrix}_{3\times 2} \mathrm{Rep}_B(x)$. Why is this the case?

5.5. We can substitute $\begin{bmatrix} | & | \\ a_1 & a_2 \\ | & | \end{bmatrix}_{3\times 2}$ with Rep_____,_____(f) as it performs a linear mapping _____ to the vector with the basis _____ representation and changes it to the _____ basis representation.

6. Let $\mathbf{x} = \begin{bmatrix} 1 \\ 2 \\ 4 \end{bmatrix}$, find $\mathrm{Rep}_B(x)$ for each basis B. (No need to find B^{-1})

6.1. $B = \left\langle \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} \right\rangle$

6.2. $B = \left\langle \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\rangle$

7. Consider a linear transformation $T\colon \mathbf{R}^2 \Rightarrow \mathbf{R}^3$ which has $\mathbf{A}_{3\times 2}$ as a matrix representation. If we write a matrix multiplication which is equivalent to $\mathbf{Ax}$ where $\mathbf{x} \in \mathbf{R}^2$, but performing with these following steps:

   1) Convert x from the standard basis B.
   2) Perform T on the result got from 1) and convert its basis to D.
   3) Convert the result got from 2) back to the standard basis.

   After performing the steps above, we got the result of $\mathbf{Ax}$ as following:

$$T(\mathbf{x}) = \mathbf{Ax} = \begin{bmatrix} 1 & -1 & 1 \\ 1 & 0 & 3 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 3 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 1 & -1 \end{bmatrix} \mathbf{x}$$

   Find the basis B and D.

8. Consider a linear transformation $T\colon \mathbf{R}^4 \Rightarrow \mathbf{R}^3$ where $T(a,b,c,d) = (a+b, b+c, c-d)$.

   Given $B = \left\langle \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \\ 2 \end{bmatrix} \right\rangle$, $D = \left\langle \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \right\rangle$ be the bases.

   Find the matrix that represents $\mathrm{Rep}_{\varepsilon,D}(T)$ and $\mathrm{Rep}_{B,D}(T)$ (No need to calculate the actual value.)

9. Consider linear transformations T: $\mathbf{R}^2 \Rightarrow \mathbf{R}^3$ and S: $\mathbf{R}^3 \Rightarrow \mathbf{R}^5$ which has $\mathbf{A}_{3\times 2}$ and $\mathbf{G}_{5\times 3}$ as matrix representations, respectively. Given B and J be the bases of $\mathbf{R}^2$ and $\mathbf{R}^5$, respectively.
(For the following questions, just leave the answer in a term of matrix multiplication.)

9.1. Find $\mathrm{Rep}_{B,\varepsilon}(T)$ and $\mathrm{Rep}_{\varepsilon,J}(\mathrm{S})$

9.2. Find $\mathrm{Rep}_{B,J}(T \circ \mathrm{S})$

9.3. If we don't know the actual value of $\mathbf{G}$, but we have D as the bases of $\mathbf{R}^3$ and $\mathbf{Y} = \mathrm{Rep}_{D,J}(S)$ instead. Find $\mathrm{Rep}_{\varepsilon,J}(T \circ \mathrm{S})$.